Samuel Howard

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EDUCATION

PhD in Statistics, University of Oxford, UK 2023–

- Thesis Title: *Controlled Diffusion Processes for Dynamic Transport of Measures*
- Supervised by George Deligiannidis and Patrick Rebeschini
- Member of the Modern Statistics and Statistical Machine Learning (StatML) CDT Programme

Master of Mathematics, New College, University of Oxford, UK 2018–2022

- Part C Results (Fourth Year Examinations): Distinction (3rd in Year Group, average score 86)
- Part A and Part B Results (Second and Third Year Examinations): First Class
- First Year Examination Results: Distinction

AWARDS

Spotlight Presentation, ICML: *Control Consistency Losses for Diffusion Bridges* 2026

Oral Presentation, FPI@NeurIPS Workshop: *Control Consistency Losses for Diffusion Bridges* 2025

Junior Mathematical Prize: Placed 3rd in cohort in Oxford Mathematics Part C Examinations 2022

Boyer Prize, New College: Best performance in Second Year Mathematics Examinations 2020

Karen Thornton Memorial Prize, New College: Best performance in First Year Mathematics Examinations 2019

Head Boy, Stockport Grammar School: Elected by both peers and staff 2018

IBM Prize, National Cipher Challenge: Achieving 1st place out of over 3,500 entries 2017

RESEARCH

Control Consistency Losses for Diffusion Bridges, *ICML 2026 (Spotlight)*
*Samuel Howard**, Nikolas Nüsken, Jakiw Pidstrigach

- Introduces a self-consistency objective for learning the conditioned dynamics of diffusion processes
- Earlier version selected as Oral presentation at Frontiers in Probabilistic Inference Workshop, NeurIPS 2025

A Mean-Field Framework for Inference-Time Distribution Control of Diffusion Models, *SPIGM Workshop, ICML 2026*
*Samuel Howard**, Nikolas Nüsken

- Develops an inference-time diffusion model steering procedure for distribution-level reward functions

Reward-Aligning Few-Step Flow Models with Integrated Regularizers, *SPIGM Workshop, ICML 2026*
Peter Potaptchik*, *Samuel Howard**, Michael S. Albergo

- Introduces a novel approach for finetuning few-step flow models to target a reward-tilted distribution

Schrödinger Bridge Matching for Tree-Structured Costs and Entropic Wasserstein Barycentres, *NeurIPS 2025*
*Samuel Howard**, Peter Potaptchik, George Deligiannidis

- Develops a novel approach for computing Wasserstein barycentres using flow-matching techniques
- Blog post: https://samuelhoward.co.uk/post/imfbarycentres

Diffusion Models and the Manifold Hypothesis: Log-Domain Smoothing is Geometry Adaptive, *NeurIPS 2025*
Tyler Farghly*, Peter Potaptchik*, *Samuel Howard**, George Deligiannidis, Jakiw Pidstrigach

- Examines how smoothing of the empirical score function influences diffusion model generalisation

Differentiable Cost-Parameterized Monge Map Estimators, *Diff. Almost Everything Workshop, ICML 2024*
*Samuel Howard**, George Deligiannidis, Patrick Rebeschini, James Thornton

- Constructs a differentiable optimal transport map estimator for training a neurally-parameterized transport cost

INTERESTS

Tutor	Tutor in Statistics, Probability at New College, Oxford (First Year Mathematics courses)	2023–
	Tutor in Integration at New College, Oxford (Second Year Mathematics course)	2023–
Music	Achieved Grade 8 with Distinction on Classical Guitar and Double Bass	2017, 2018
	Member of Oxford University Philharmonia (2018–2024), Hallé Youth Orchestra (2017–2018)	
Other	Gold Duke of Edinburgh's Award (2018)	